Characterizing Indel Mutations in *Arabidopsis thaliana* as a Model For Compensatory Evolution

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Introduction

Limited research exists regarding the phenomenon of compensatory evolution—when the genome resolves deleterious mutations potentially increasing the fitness of the individual. This study examined patterns and characteristics of insertion and deletion (indel) mutations within the genome of *Arabidopsis thaliana* in order to parameterize a model of compensatory evolution.

Codons:
TAT-GCT-TAT-CCT-CTG-GCT-AAT-ACT-TCT



ABBREVIATIONS:
Y-A-Y-P-L-A-N-T-S

Insertion within 7th codon:
TAT-GCT-TAT-CCT-CTG-GCC-TAA-TACACTTCT…

ABBREVIATIONS:
Y-A-Y-P-L-A-*A-stop*

This insertion causes a premature stop codon (PSC) and therefore a frameshift mutation at that point.

Compensatory mutation:
TAT-GCT-TAT-CCT-CTG-GCC-TTT-AAT-ACT-TCT

ABBREVIATIONS:
Y-A-Y-P-L-A-*F*-N-T-S

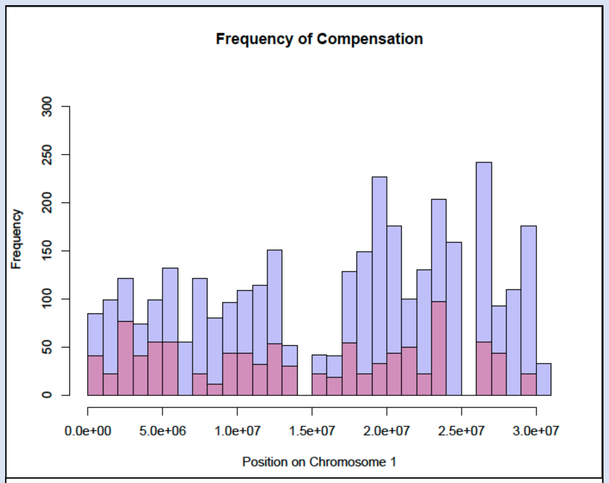
The insertion of two thymine restores frame and is potentially less deleterious than the PSC.

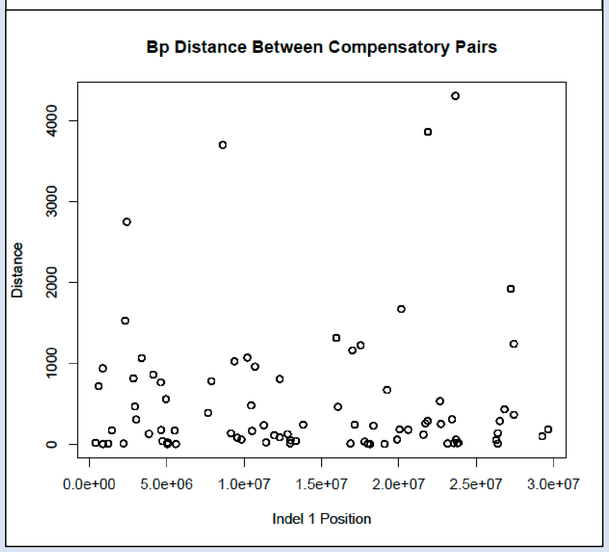
Parameters

The sample used in this study represents:
- 180 *A. thaliana* samples
- 500k short indel reads

Examined characteristics such as:
- Indel position
- Position of compensatory pairs
- Number of base pairs between these compensatory pairs

Initial Findings





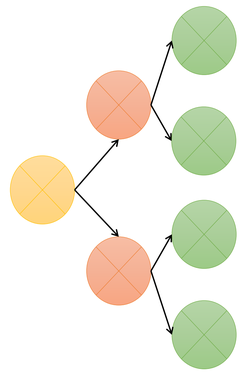
The figures above represent the correlation between the position of the indel mutation and the frequency of compensation. These graphs are representative of all genes created by loci pairs in 10 individuals. It can be noted that the frequency of compensation appears to be less dependent on the position and more correlated to the distance between the compensatory indel. This relationship appears to be inversely proportional.

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Mathematical Modeling

- Assumed to be a Markovian process
- Branching process
- Parameterized by patterns identified
- Robust to various genomes

Branching Process



This graphic symbolizes the branching process purposed to model this phenomena. Each node represents a different characteristic of the mutation and as the process progresses, the rate of compensation can be determined.

Implications

"There is no branch of mathematics, however abstract, which may not some day be applied to phenomena of the real world." -Nikolai Ivanovich Lobachevsky

- Model fitting to various genomes
- If an indel is the problem, when is it the solution?
- Applicable in biomedical, pharmaceutical, and agricultural sciences

Project Outlook

- Further parameterize model based on current findings
- Create model robust to various genomes and sample sets
- Analyze possible mechanisms of compensatory evolution
- Understand compensatory evolution on the biochemical scale

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